Zhaodong Liu

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EDUCATION

New York University Shanghai — Shanghai, China

Bachelor of Science; **Double Major**: Computer Science, Mathematics — Sep 2022-May 2026 (expected)

Cumulative GPA: **3.811/4.0** | CS Major GPA: **3.941/4.0** | Math Major GPA: **3.926/4.0**

- **Study-away Experience:** NYU New York, Fall 2024 and Spring 2025
- **Relevant Coursework:** Multivariable Calculus, Honors Linear Algebra, Probability & Statistics, Honors Numerical Analysis, Discrete Mathematics, Analysis, Introduction to Computer Science & Data Science, Data Structures, Basic Algorithms, Introduction to Databases, Natural Language Processing, Java and Web Design

RESEARCH EXPERIENCE

Research Assistant, *New York University Shanghai* — May 2025-present

Generative Retrieval for Recommendation System

Advisor: Prof. Hongyi Wen, Department of Computer Science, Data Science and Engineering

- Developed models to predict user behavioral patterns on Amazon datasets.
- Designed and evaluated model architectures, including RQ-VAE, VQ-VAE, and OPQ, to capture discrete behavior patterns, achieving 47.2% performance improvement (from 0.0339 to 0.0499).
- Implemented advanced tokenization methods, including BPE-inspired partitioning, improving sequence coherence and representation quality.
- Adapted the model framework to support multimodal representation learning.

Research Assistant, *New York University Shanghai* — May 2024-Aug 2024

Efficient Driving Route Optimization Considering Traffic Light Timings

Advisor: Prof. Zhibin Chen, Department of Computer Science, Data Science and Engineering

- Built a traffic-aware route optimization system under real-world constraints.
- Applied Dijkstra's algorithm and related methods to compute the optimal path(s).
- Designed distinct power-consumption dynamics models for gasoline vehicles and electric vehicles.
- Simulated model performance on real traffic signal datasets, demonstrating significant reduction in travel time compared to baseline routing algorithms, while maintaining efficiency.

SELECTED PROJECTS

Finetuning on Pretrained LM to Efficiently Perform Retrosynthesis of Chemical Reactions Sep 2024-Dec 2024

Center for Data Science, New York University

- Developed predictive models for chemical reaction retrosynthesis using SMILES notation.
- Applied an LLM for dataset reconstruction and augmentation, improving Top-1 Accuracy by 26.2% (from 0.588 to 0.742) while reducing reliance on expensive laboratory data.
- Employed parameter efficient fine-tuning strategies (encoder freezing, MLP adapters, LoRA) to optimize training efficiency.

Music Snippet Classification based on Vocal Gender Types — Mar 2024-May 2024

New York University Shanghai

- Built an audio classification pipeline to categorize music snippets according to vocal gender types.
- Implemented mel-frequency feature extraction and vocal separation to isolate singer characteristics.
- Designed and trained ResNet34 models, adapted vision architectures for audio spectrograms.
- Implemented an ensemble of 20 models, improving the performance by 2–3% over single-model baselines, achieving 80.42% (top 10%) accuracy.

WORK EXPERIENCES

Data Engineer, Data Strategy Department, *McDonald's China* — Dec 2025-present

- Designed and validated scalable database schemas and enterprise-level data maps.
- Built a similarity recommendation algorithm based on menu composition and user historical interactions.
- Implemented an AI agent that significantly enhanced data panel classification and visualization.

LEADERSHIP & ACTIVITIES

Treasurer, PEER-Pal Club, *New York University Shanghai* — May 2023-Aug 2024

- Managed financial operations and annual budgeting for student-led educational initiatives.
- Organized exhibitions and club fairs, strengthening engagement with the campus community.
- Facilitated letter exchanges and Q&A sessions with high school students from Hunan Province.

Trombone Section Leader, Chamber Orchestra, *New York University Shanghai* Sep 2022-present

- Led the trombone section, coordinated rehearsals, and oversaw instrument storage logistics.

DISTINCTIONS

Mathematical Contest in Modeling, Certificate of Achievement, Honorable Mention May 2023

- Preprocessed large datasets to remove anomalies and ensure data quality.
- Built predictive models using methods including Bayesian regression for market analysis.
- Synthesized regional market data to support modeling results.

Dean's List, *New York University Shanghai* Academic Years: 2022-2025

- Recognized for outstanding academic performance

SKILLS AND OTHERS

- **Programming:** Python, Java, C, SQL, Stata
- **Frameworks & Tools**: Git, PyTorch, Pandas, NumPy, LaTeX
- **Languages**: Chinese (Native), English (Fluent)